

Every options agent reports what it made. None report what it paid.

An autonomous options agent on Alpaca paper trading that measures execution cost before it places an order, and structures every position so it never pays that cost a second time.

A CONTRACT QUOTED AT 15:34 ET

BID 0.00 ASK 0.08

IWM put, 26 minutes to expiry. It is worth four cents and unsellable.

EXIT COST PAID TO DATE

\$0.00

PUBLISHED CLAIMS REPRODUCING

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Alpaca paper account **PA31RB6YR1V6**, opened at exactly \$100,000 and never traded by hand.

ahammadshawki8 · github.com/ahammadshawki8/halfspread · ahammadshawki8.github.io/halfspread

The cost of leaving a trade is not a rounding error. Near expiry it is the trade.

Half the bid-ask, as a share of what the contract is worth. Thirteen names quoted within the same minute on our own broker feed. The only thing separating the two blocks is how long the contract has left to live.

TICKER	TIME LEFT	HALF-SPREAD
NVDA	a day	0.52%
TSLA	a day	1.16%
AAPL	a day	1.27%
MSFT	a day	2.95%
GOOGL	a day	4.97%
SPXW	26 minutes	10.20%
SPY	26 minutes	14.29%
QQQ	26 minutes	33.33%
IWM	26 minutes	100.00%

A strategy screen that picks strikes on delta lands exactly where this curve is worst. **Our agent picks on expected value after the measured cost instead.**

Measured 3 Sep 2026 on the Alpaca indicative feed. These are the costs this account was charged, not a claim about the true OPRA spread.

Entry and exit cost the same at any one instant. So "the exit is expensive" is a claim about time.

We re-quoted the same contracts every few minutes for a full session to test it. Below is a position the desk actually held, priced at the opening bell and again near the close.

SPY 765 PUT, AT THE OPEN

1.235%

bid 0.40 / ask 0.41, on a mid of 0.405

THE SAME CONTRACT, 19:56 UTC

2.857%

bid 0.17 / ask 0.18, on a mid of 0.175

COST TO LEAVE, RELATIVE

2.31x

The gap held at a penny while the option decayed, so the same penny became twice the price.

Why it compounds

Two forces push the same way: dealers widen quotes as gamma risk rises into the close, and the contract itself decays. A fixed spread on a shrinking premium is a rising percentage.

Who pays it

Every agent that opens a position in the morning and closes it in the afternoon. That is the standard design, and it pays the widest spread of the day on the way out.

The way out of it

An out-of-the-money contract that reaches expiry is removed at zero. No closing trade exists, so no spread is crossed and no commission is charged.

89 observations of this contract are in the journal, timestamped, and the chart on the live site is drawn from them. **The exit is where the money dies, and it dies on a clock.**

Build the agent backwards from the cost, and the exit disappears.

HALFSPREAD sells defined-risk option spreads and chooses strikes with one objective: that the contracts expire worthless. The structure is ordinary. The selection criterion and the intent to hold are not.

1 • Observe Chain, quotes and underlying across SPY, XSP and SPXW every cycle	2 • Price Own Black-Scholes, own greeks, own implied volatility from the mid	3 • Measure cost Half-spread per leg from live quotes, before an order exists	4 • Net EV Rank on value after that cost, not on a delta target	5 • Gate Win probability, loss caps, daily budget, concurrency	6 • Settle Hold to expiry. No closing trade to pay for
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CANDIDATES PRICED

40,629

across 235 scans

PASSED EVERY GATE

958

2.4% admitted

REFUSALS JOURNALLED

37

each with its arithmetic

EXIT COST, BY DESIGN

\$0.00

settlement is not a trade

Both legs are covered inside a single Alpaca **mleg** order, so the loss is capped before the position exists and there is no naked short anywhere in the book.

A leading 2026 study was corrected last month for exactly this error. Every result flipped.

Vilkov's ODTE trading-rules paper charged the bid-ask half-spread at one hundredth of its true size, 0.022 basis points instead of 2.2. External replicators found it. The corrected net Sharpe ratios:

STRUCTURE, NEAR THE MONEY	BEFORE	AFTER
Put ratio spread	+0.84	-0.61
Risk reversal	+0.44	+0.10
Bear put spread	+0.30	-0.73
Strangle / straddle	-0.51	-0.97
Iron butterfly / condor	-0.96	-2.67
Top-3 conditional basket	+0.82	-0.82

- No strategy or basket retains a positive net Sharpe ratio.*
- The author's own conclusion, after the correction.
- **We do not claim his numbers support us.** His structures are near the money and round-tripped at 16:00. Ours are further out and held to settlement. Different regime, separate evidence.
 - **We cite him for the cost finding,** which is the part that generalises: mis-price execution and every conclusion downstream is wrong.
 - The annotated paper in that repository still shows pre-correction figures. Only the issues file is current.

Supporting evidence we do rely on: a 40-delta 7DTE bull put spread held to expiration ran an **80% win rate from 2023 to mid-2026**, and Alpaca charges no commission on contracts that simply expire.

We surveyed roughly ninety submissions before writing a line of code.

Not to copy them. To find out which claims would be worth nothing by the time a judge reached the ninetieth deck, and to make sure our differentiator was genuinely unoccupied.

TERRITORY	ENTRIES	VERDICT
LLM proposes, gates decide	~30	saturated
Credit spreads / premium selling	~12	crowded
Multi-agent bull / bear debate	~10	crowded
News and sentiment	~8	crowded
Backtest-first research agents	~5	occupied
Execution cost and slippage	~4	contested
Dealer gamma, earnings crush, stat arb	~5	occupied
Index options, cash settlement	0	empty

WHERE WE ACTUALLY DIFFER

- **Exit-cost avoidance as the design principle**, not a cost model bolted onto a signal. One rival states it closes every position before expiry, which is the literal anti-pattern.
- **The Vilkov correction**. A reversal of a leading paper, caused by precisely the error the field is making. No other entry has it.
- **Index options and cash settlement** as the mechanism that makes a zero-cost exit possible. Zero competitors touch them.
- **The CLI as the execution path**. Most entries used MCP or the raw SDK. Every order we place is a replayable command.

The strategies that Vilkov's correction invalidated are the ones roughly a dozen entries here are still running. **Being right about cost is not a nice detail. It decides whether the rest of the work counts.**

A deterministic core, a narrow AI, and a journal that everything is derived from.

DECISION LAYER

- **pricing.py** Black-Scholes, implied volatility by bisection, greeks
- **chain.py** OCC parsing, chain fetch, put-call parity forward
- **cost.py** half-spread measurement, net EV, admissibility
- **strategy.py** candidate construction and ranking
- **risk.py** shortfall sizing, loss caps, concurrency

RUNTIME

- **execute.py** mleg build, re-quote, submit through the CLI
- **monitor.py** net-of-cost exit test, emergency close
- **settle.py** expiry accounting and the counterfactual
- **llm.py** bounded event-risk review, clamped in code
- **session.py** one entrypoint, local or GitHub Actions

EVIDENCE

- **journal.py** append-only JSONL, fsynced, one line per decision including refusals
- **verify.py** re-derives every published claim with no key and no network
- **publish.py** builds the dashboard payload from the journal alone
- **mcp_readonly.py** stdio proxy that strips mutating tools

Zero dependencies requirements.txt is empty. Standard library only, so a judge can clone and run it with nothing installed.	GitHub Actions is the runtime A scheduled job runs a session and commits its own journal, so the audit trail is timestamped by git rather than by the agent.	GitHub Pages is the front end The page computes nothing. It renders a payload derived from the journal, so it cannot drift from the record.	No card, no paid tier Free Alpaca data, free Groq inference, free hosting. Every constraint in this project was solved without spending anything.
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7,060 journal records so far. **The journal is the product; the dashboard is only a view of it.**

The free data feed gives no greeks and no spot price. So we compute the pricing layer ourselves.

This began as a blocker and became the most defensible part of the build. Strike selection rests on mathematics we own and can show, rather than on a vendor field we would have to trust.

Greeks are all zero Every contract at every expiry returns delta, gamma, theta and vega of exactly 0, with implied volatility null. Not a market-hours artifact. Solved: implied volatility recovered from the mid by bisection, then delta, gamma, theta and vega derived from it.	No spot for cash indices Alpaca publishes no underlying price for XSP or SPXW, so moneyness cannot be computed the obvious way. Solved: the forward is recovered from put-call parity on the chain itself, using calls and puts at a shared strike.	Real markets are not lognormal Black-Scholes understates the left tail, which is precisely the tail a put seller is exposed to. Solved: breach frequency counted from 5,539 hourly bars across 668 sessions, bucketed by hour. Sizing takes whichever estimate is worse.
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MINIMUM WIN PROBABILITY	MAX LOSS PER POSITION	MAX LOSS PER DAY	VARIANCE PREMIUM ASSUMED
0.85	\$700	\$1,400	12%

The last figure is the honest one: the variance risk premium haircut is **an assumption, not an observation**, and it is where our modelled edge comes from. We say so on the site rather than burying it.

Sized against the tail that actually happened, not the one the model prefers.

Return-to-close distribution for SPY, measured by entry hour. A position opened at 09:30 faces a materially wider tail than the same position opened at 14:00, and the agent sizes accordingly.

ENTRY HOUR, ET	1ST PCT	5TH PCT	STD DEV	SESSIONS	LEFT TAIL TO THE CLOSE
09:30	-1.98%	-1.04%	0.76%	668	
10:00	-1.85%	-0.85%	0.70%	668	
11:00	-1.54%	-0.71%	0.63%	668	
12:00	-1.36%	-0.63%	0.55%	668	
13:00	-1.16%	-0.56%	0.39%	662	
14:00	-0.94%	-0.46%	0.32%	662	

The sizing rule
Breach probability is computed two ways, from the model and from this table. The agent always sizes on the worse of the two.

Defined risk, always
Both legs covered in one order. Maximum loss is known before submission and capped per position and per day.

Survivability over headline
Credit to width runs 5 to 9%, so a large dollar P&L would need 6 to 10% of the account at risk. We refuse that trade.

5,539 hourly bars since January 2024, recomputed from the broker's own history. **An 80% win rate still loses one day in five, and the sizing assumes it will.**

There is a language model in here. It is allowed to do exactly one thing: make a position smaller.

Roughly thirty entries in this hackathon pitch "the LLM proposes and risk gates decide". That territory is saturated, and we think it is also the wrong shape. Our model has authority over one scalar.

Deterministic engine Strike, width, size, limit price and exit are all arithmetic. Decided before the model is consulted, and unchanged if it never answers.	AI event review Reads live headlines through Alpaca's news API. Returns one number.	Clamped in code Forced into the range 0 to 1. A model that asks to double gets truncated to no change.	Final size Scaled down, or unchanged. Never up.
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THE MODEL CANNOT

- Choose a strike, a width or an expiry
- Enlarge a position by any amount
- Overrule a risk gate or a loss cap
- Abstain from trading entirely. A block is converted to a floor of 0.25 and logged as a conversion, because refusing a capped-loss trade over tail risk is the wrong response to a bounded tail.

WHAT IT EARNED ITS PLACE DOING

- On its first live session it pulled Strait of Hormuz attacks and a crude oil surge out of the headlines and cut the position.
- It independently flagged the payrolls print scheduled for the following morning.
- Reading unstructured event risk off the wire is the one job in this pipeline a human is genuinely worse at automating.
- If Groq is unreachable, the deterministic decision stands and the desk keeps trading.

Four vetoes are recorded in the journal with their reasoning and their resulting multiplier. **Give the model less to do, and what it does can be trusted.**

The CLI is the agent's hands. MCP is the window. Read-only is enforced, not promised.

EXECUTION, THROUGH THE CLI

- Every broker action is a logged **alpaca** invocation, recorded verbatim in the journal and replayable by hand from the command line.
- Multi-leg orders go out as **order_class: mleg** with both legs covered inside the same order, which is what makes them Level 3 legal with no naked short.
- **A finding worth the deck space:** Alpaca signs an mleg limit price from the package's point of view. Positive is a maximum debit, negative is a minimum credit. A credit spread submitted positive silently behaves as a market order. We proved it three ways and the sign is now forced in code.
- Decision-to-execution drift measured at **\$7 per contract** against a \$1 bid-ask, so every order re-reads both legs seconds before submission and journals the drift.

INSPECTION, THROUGH MCP

- Alpaca's official MCP server, v3.4.7, wired into the project and verified over stdio.
- The vendor's own toolset filter does not remove **place_option_order**, so a configuration promise would have been worth nothing.
- We wrote a stdio proxy that sits in front of it, strips mutating tools from the listing and refuses them by name on call.

TOOLS BLOCKED

11

orders, cancels, closes, exercises, account config

TOOLS ALLOWED

24

read-only account, positions, market data, news

Auditable in one command: `python -m agent.mcp_readonly --audit`

A judge can point their own MCP client at this repository and inspect the live account without being able to move a single dollar. **The setup is documented in the README and on the site.**

Even the emergency exit is decided on net cost, never on a distance trigger.

The obvious rule is to close when the underlying approaches the short strike. For a defined-risk spread that rule is actively harmful, and it is worth explaining why, because it is where most agents lose money.

WHY THE OBVIOUS RULE IS WRONG

- The loss is **already capped** by the long leg. There is no tail to escape.
- By the time a short strike is threatened, the exit spread has widened to several times entry cost. We measured **3.92x on SPY in a single session**.
- So the trigger pays the exact cost this agent exists to avoid, in order to escape a loss that was bounded from the beginning.

THE TEST WE USE INSTEAD

What does buying the package back cost **right now**, against the expected terminal intrinsic if we simply hold? Close only when the market is charging materially less than we expect to pay at settlement.

Materiality is the greater of **\$25** or **25%** of the exit cost, which stops the desk churning on modelled edge that sits inside the noise of our own volatility estimate.

It is the same net-of-cost test the entry uses, pointed the other way.

Holding is a decision too When the desk holds through a breach it writes a hold_through_breach record with the full arithmetic, so a decision not to act is as visible as one to act.	Emergency close still exists It is booked explicitly as a cost event rather than hidden in the P&L, so the ledger never flatters itself.	Used so far Zero times. Every position resolved at settlement, which is what the strategy is for.
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Assignment risk on an ITM SPY finish is real and disclosed. **It only bites when the short strike is already breached, which is the one case the arithmetic above is written to handle.**

Two positions settled. The interesting number is not the profit.

POSITION	CONTRACTS	CREDIT	UNDERLYING AT SETTLEMENT	OUTCOME	REALISED
SPY 765 / 763 put spread	3	\$30.00	773.23	expired OTM	+\$39.00
XSP 767 / 764 put spread	2	\$39.00	774.78	expired OTM	+\$32.00

REALISED P&L

+\$71.00

2 of 2 positions won

ENTRY COST PAID

\$7.00

measured before each order

EXIT COST PAID

\$0.00

no closing trade existed

IF ROUND-TRIPPED INSTEAD

+\$61.00

the same two trades, closed

The counterfactual is the result

Not paying to exit was worth \$10.00, or **14% of the gain**, on two small positions in a single session. That ratio is the thesis, and it scales.

Why the number is small

Credit to width runs 5 to 9%. Sizing for survivability caps a clean day near +0.1 to 0.3%. We did not crank risk to buy a headline.

Disclosed, not smoothed

Broker equity reads \$100,032.65 because Alpaca clears expiries overnight. The site shows both figures and explains the gap.

3 orders filled, \$69.00 of credit received, 0 emergency closes, 1 position still open into Friday. **A red account would poison every other criterion, so the posture is a high win-rate grind.**

Our dashboard once published \$316 that never happened. Here is what we did about it.

The ledger showed \$316 realised across six settled positions while every contract still had six hours to run. Two bugs: settlement compared only the expiry date, and journal records were not scoped by account, so DEV experiments were being attributed to the competition account.

What we did not do Quietly delete the records and re-publish. That is the version nobody would have caught.	What we did Removed the eleven fabricated records and appended a journal_correction in their place, so the removal is a permanent part of the audit trail rather than an absence in it.	What it produced A verifier that re-derives every published claim from the committed journal, and regression tests for both bugs. Neither can reach production again without failing the build.
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CLAIMS RE-DERIVED

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python -m agent.verify

TESTS, CREDENTIAL-FREE

45

runs on every push in CI

JOURNAL RECORDS

7,060

append-only, committed to git

NEEDED TO CHECK US

0

no key, no account, no network

Every claim in this deck is in that verifier, and the exit code is non-zero if any of them stops reproducing. **When a dashboard is generated from a journal, a journal bug is a published lie.**

Audited against the 7-dimension reproducibility checklist in arXiv:2606.08285. Costs are Alpaca indicative-feed costs, not OPRA, and we say so everywhere the figure appears.

Sixty seconds is enough to check everything above.

THREE THINGS WORTH DOING, IN ORDER

- **Open the site and price a trade.** The page ships the live chain, the gate thresholds and the measured breach distribution, so you can invent a spread and the same engine that decides our orders will tell you what it costs, what leaving would cost, and which gate rejects it. Most get rejected. That is the point of it.
- **Run the verifier.** Clone the repo and run one command. It re-derives every number in this deck from the committed journal with no key, no account and no network.
- **Point your MCP client at the account.** Setup is in the README. You will get 24 read-only tools and be unable to move a dollar.

LINKS

ahammadshawki8.github.io/halfspread
github.com/ahammadshawki8/halfspread

COMPETITION ACCOUNT

PA31RB6YR1V6
0509a308-f1ef-44e2-8e8b-8a6d0893f84b

BUILT WITH

Alpaca Trading API, Alpaca CLI, Alpaca Market Data, Alpaca MCP Server, Python standard library, Groq, GitHub Actions, GitHub Pages.

Most agents will tell you what they made. This one can show you what it paid.